The Bridge Study — Supplement (v3.1)

Jack McCarthy
WikiLean · https://wikilean.jackmccarthy.org

Companion to the **Version 3.1** report — August 18, 2026

Companion to `docs/research/BRIDGE-REPORT.md` (version 3, 2026-08-02; typeset as `report.pdf`). Everything the main paper's structure moved out lives here: the complete preregistration execution inventory, raw-instrument tables and outage bases, instrument-validation detail, exposure strata, sensitivity analyses, the full tool-call census, the WF manual, review responses, the changelog, and data availability. Section numbers §Sn are cited from the main paper. This supplement also carries the exact asymmetric interval bounds — [lo, hi] — behind every "± (95% confidence)" value the main paper quotes. Nothing here is new evidence; every table is recomputable from the file named beside it.

S1 Complete preregistration execution inventory

Table S1 lists every preregistered component, its status, and the inferential consequence. Its sources are the preregistration (`docs/research/BRIDGE-EXPERIMENT.md`, commit `0d36f266`) and the claim-by-claim inventory in `docs/research/review/verification-of-review-1.json` (prereg section).

Table S1: Preregistration execution inventory.

Preregistered component	Status	What actually happened	Inferential consequence
Five arms A–E, identical model/prompt/budget, tools-only difference	executed	471 rows/arm (`bench/data/runs/`)	design intact
Per-tool call logging	executed	`tool_calls_by_name` every row; full traces eval C/D/E + fresh (deviation 7)	enables trace attribution
Tier 1a ProofNet# (371)	executed	371/arm, scored	contaminated by construction (arm A 59.8% on eval-341); paired deltas only
Tier 1b fresh set (∼100)	executed	100 tasks	holdout narrower than claimed (Brain indexes only; one later-fold leak) — renamed post-Brain-index, 99/100
Hallucinated-decl rate	executed	all tables	instrument required repair (§S3); repaired rates are the report's
McNemar D-vs-E / D-vs-C + discordant tables	executed	main §4	run on grounded typecheck, not the preregistered endpoint
10-min wall clock	executed	600 s timeouts	—
Pre-campaign API requirements 1–8	executed	implemented before campaign	—
30-turn budget	**modified**	advisory only; overruns C 50/D 38/E 48 on the repaired rows (E 32 as-run — outage rows masked overruns), max 88	budget uncontrolled; §S5 sensitivity
Determinacy pre-screen (exclude)	**modified**	post-hoc dual annotation (79%, $\kappa \approx 0.20$); 74-task subset, none excluded	subset analysis, not screened population
Arm-D production server	**modified**	production Worker code served locally over shipped shards	none material
Single-shot elaboration grading	**modified**	persistent REPL, same pins	none material
≤4 typecheck calls in-loop	**modified**	no in-loop typechecking at all	halluc rate is a pure grounding measure
Skill/ToolSearch leak	**modified**	sealed before eval; 120 dev runs quarantined	dev split discarded, eval clean
Per-arm cost distributions	**modified**	means only (USD + wall-clock)	descriptive (§S10)

Preregistered component	Status	What actually happened	Inferential consequence
faithful@budget (BEq+ equivalence)	**not executed**	grader a stub; zero judge files in campaign	primary endpoint missing; zero confirmatory results
LLM judge + 50-item human calibration	**not executed** (campaign)	harness existed unused; an *uncalibrated* blind judge pass ran post-review (§S11)	main §4.2 is exploratory only
tokens-to-solve (half the success criterion)	**not executed**	never computed (tokens recorded per row)	P2 untested as specified
3 reseeds + pass@k curves	**not executed**	one run per (task, arm) everywhere	no variance estimate anywhere
Second model class on primary set	**not executed**	Tier-1 all-Haiku; v2 all-Sonnet on different benchmarks	capability-band generality unknown
Preregistered success criterion	**not executed**	neither half graded as specified	zero confirmatory results
Tier 2 as specified (FATE-H + MPR-Prop proving, reflection loop)	**not executed**	replaced by exploratory v2: QR/MPR retrieval + one-shot SorryDB, arms N/F/W/U/WF	main §§5–6 exploratory
PutnamBench	**not executed**	—	—
Tier 3a offline Erdős set	**not executed**	—	—
Tier 3b live Erdős queue	**not executed**	—	—

S2 Tier-1: raw-instrument tables and analysis bases

Sources for this section, all under `bench/analysis/`, are `tier1_reanalysis.{py,json,md}`, `part1_fresh100_v2.{json,md}`, `fresh_clustered.{py,json,md}`, and `success_repaired.{py,json,md}`. Grading pins: eval-341 on Lean v4.32.0-rc1 / Mathlib `a33a5ccd`; fresh-100 on Lean v4.33.0-rc1 / Mathlib `9944fe29`; the checkout C/E's file tools read is `61a5e4f338` (content ∼2026-07-10); ProofNet# source pin `PAug/ProofNetSharp @ a8da405f` (MIT).

S2.1 The three arm-E bases (raw instrument)

E's 31 infrastructure-dead rows (fresh_069–099, contiguous session-limit 429s; A–D verified zero errors) can be counted three ways. Cells show grounded typecheck as n/N = % [Wilson 95% CI], on the **raw oracle**:

arm	errors-as-failures (n=100)	completed pairs (n=69)	post-repair (n=100)
A	20/100 = 20.0% [13.3, 28.9]	10/69 = 14.5% [8.1, 24.7]	20/100 = 20.0% [13.3, 28.9]
B	22/100 = 22.0% [15.0, 31.1]	12/69 = 17.4% [10.2, 28.0]	22/100 = 22.0% [15.0, 31.1]
C	25/100 = 25.0% [17.5, 34.3]	12/69 = 17.4% [10.2, 28.0]	25/100 = 25.0% [17.5, 34.3]
D	42/100 = 42.0% [32.8, 51.8]	28/69 = 40.6% [29.8, 52.4]	42/100 = 42.0% [32.8, 51.8]
E	16/100 = 16.0% [10.1, 24.4] — 31 infra errors	16/69 = 23.2% [14.8, 34.4]	30/100 = 30.0% [21.9, 39.6]

Raw McNemar (exact binomial two-sided): errors-as-failures D-vs-E 32/6 p=2.4e-5 (conflates outage with behavior — quarantined here); completed-69 D-vs-E 18/6 p=0.023, D-vs-C 21/5 p=0.0025, D-vs-A 22/4 p=0.0005; post-repair full-100 D-vs-E 25/13 p=0.073, D-vs-C 28/11 p=0.0095, D-vs-A 29/7 p=0.0003, E-vs-A 19/9 p=0.087. Of the 31 repaired E rows, 14 became successes.

Wald/McNemar mismatch note. On the post-repair full-100 raw table, the paired-Wald D−E risk-difference interval was [+0.0015, +0.2385] (barely excluding 0) while exact McNemar gave p=0.073 — a normal approximation over all 100 paired differences versus a test conditioned on the 38 discordant pairs, landing on opposite sides of α=.05 near the boundary. This is why v3 replaces both with a single commit-clustered bootstrap whose interval and p-value are read off the same resampling distribution.

S2.2 Raw vs repaired instrument, side by side

Repaired oracle = `halluc_validation.py classify_adjusted` (rules R1–R5, §S3), folded into success by `success_repaired.py`; success is monotone under the repair (flags are only ever cleared). Affected rows (raw-flagged, repaired-clean) and how many then typecheck:

arm	affected (fresh)	typecheck pass	affected (eval-341)
A	17	1	39
B	18	0	45
C	38	8	68
D	17	6	47
E	39	7	81
total	129	22	280

arm	raw k/n [Wilson 95]	repaired k/n [Wilson 95]	Δ
A	20/100 [13.3, 28.9]	21/100 [14.2, 30.0]	+1
B	22/100 [15.0, 31.1]	22/100 [15.0, 31.1]	+0
C	25/100 [17.5, 34.3]	33/100 [24.6, 42.7]	+8
D	42/100 [32.8, 51.8]	48/100 [38.5, 57.7]	+6
E	30/100 [21.9, 39.6]	37/100 [28.2, 46.8]	+7

McNemar, both instruments:

pair	raw b/c	raw p	repaired b/c	repaired p	classification
D vs E	25/13	0.073	28/17	0.135	ns → ns
D vs C	28/11	0.0095	32/17	0.044	sig → sig
D vs A	29/7	0.00031	34/7	2.5e-5	sig → sig
C vs A	16/11	0.442	23/11	0.058	ns → ns
E vs A	19/9	0.087	24/8	**0.007**	**ns → sig**

Commit-clustered paired bootstrap (44 clusters, B=10,000; raw column asserted byte-identical to `fresh_clustered.json` before the repaired column was computed):

pair	raw RD [95% CI] p	repaired RD [95% CI] p	classification
D − E	+0.120 [−0.049, +0.271] p=0.192	+0.110 [−0.089, +0.279] p=0.304	ns → ns
D − C	+0.170 [+0.011, +0.314] p=0.040	+0.150 [−0.023, +0.302] p=0.100	**sig → ns**
D − A	+0.220 [+0.093, +0.336] p=0.0014	+0.270 [+0.131, +0.395] p=0.0004	sig → sig

Note the two classification changes at α=.05 cut in opposite directions (E-vs-A strengthens, D-vs-C weakens) — the repair is not a thumb on either scale.

S2.3 Cluster census and sensitivity collapses

The 100 tasks decompose into **44 distinct source commits**, **57 files**, and **59 name-stem families** (24 multi-member). The largest commit clusters are `87a6eccf` (9 tasks, the AntitoneOn sum–integral family), `49ed1b2d` (8, bounded variation), `61303857` (5), two of size 4. On the raw instrument, +7 of D's net +12 paired advantage over E comes from the 9-task commit (D 7/9, E 0/9) and +3 from the 8-task commit (D 4/8, E 1/8) — 83% of the net effect in 2 of 44 commits.

All pairs × all clustering levels (raw instrument, paired bootstrap RD [95% CI] p):

pair	task-level (iid)	family-clustered	file-clustered	commit-clustered
D − E	+0.120 [+0.000, +0.240] p=0.062	+0.120 [−0.044, +0.274] p=0.186	+0.120 [−0.044, +0.274] p=0.178	+0.120 [−0.049, +0.271] p=0.192
D − C	+0.170 [+0.050, +0.290] p=0.0064	+0.170 [+0.021, +0.311] p=0.032	+0.170 [+0.021, +0.310] p=0.030	+0.170 [+0.011, +0.314] p=0.040
D − A	+0.220 [+0.110, +0.330] p=0.0002	+0.220 [+0.080, +0.346] p=0.0028	+0.220 [+0.080, +0.344] p=0.0026	+0.220 [+0.093, +0.336] p=0.0014

Family/file/commit-collapsed sensitivity (one unit per cluster, majority and any-success rules) is tabulated in `bench/analysis/fresh_clustered.md` §4; under every collapse D−E is a null ($p \geq 0.52$), D−A survives the any-success collapses (p=0.027–0.041), and the tie-free mean collapse gives D−E +0.043 [−0.104, +0.183] p=0.554, D−A +0.157 [+0.043, +0.264] p=0.0064.

S2.4 Eval-341 (ProofNet#) raw table and the repaired-instrument gap

arm	grounded typecheck (raw)	Wilson 95% CI
A	204/341 = 59.8%	[54.5, 64.9]
B	198/341 = 58.1%	[52.8, 63.2]
C	218/341 = 63.9%	[58.7, 68.9]
D	219/341 = 64.2%	[59.0, 69.1]
E	208/341 = 61.0%	[55.7, 66.0]

The 280 raw-flagged repaired-clean eval rows (per-arm counts in S2.2) exceeded the typecheck budget and were **not** re-typechecked; the counts are hard upper bounds on per-arm gains, and the instrument-bias direction (C/E over-flagged relative to D) applies to this table too. Typecheck alone anti-correlates with grounding on this split (E led typecheck at 83.3% while trailing on hallucinations), reproducing TheoremGraph's typecheck-is-not-a-signal finding at 15× their n.

S3 Hallucination-oracle validation detail

Sources for this section are `bench/analysis/halluc_validation.{py,json,md}`, with blinded intermediates in `bench/analysis/halluc_blind/`.

Blinded protocol. The audit drew a seeded (20260801), stratified sample of 60 distinct cited names over arm × oracle-verdict strata (per arm: 6 flagged hallucinated, 5 exists, 1 renamed). The grader saw only the blinded sample (name + one context line, shuffled, no verdicts) and raw `git grep` evidence at the pinned trees (`61a5e4f338`, cross-checked at `9944fe2973`), wrote truth labels, and only then unsealed the key. Grading required namespace resolution, `to_dual`-generation tracing, `_root_.` declarations, and structure-field projections — exactly the care an exact-string oracle cannot apply.

Truth labels: real 46 · nonexistent 4 · not_a_citation 9 (extractor noise) · renamed 1.

Confusion (oracle → truth):

	truth real/renamed	truth nonexistent	truth not_a_citation
oracle hallucinated (30)	18	4	8
oracle exists (25)	24	0	1
oracle renamed (5)	5	0	0

Binary scores (positive = hallucinated): strict precision 4/30 = **13.3%** [5.3, 29.7], recall 4/4 = 100% [51.0, 100]; lenient (noise rows excluded) precision 18.2% [7.3, 38.5]. None of the oracle's 30 negative verdicts (25 *exists* + 5 *renamed*) was a missed fabrication (FN=0); read as verdicts rather than a pooled negative class, 24/25 *exists* names denoted real declarations (one was extractor noise) and 4 of the 5 *renamed* verdicts were in truth real names.

The four confirmed fabrications (zero-hit at both revs), all from arms without formal tools: `FractionField` (A; real name `FractionRing`), `Localization.At` (B; `Localization.AtPrime`), `Ideal.vanishingLocus` (B; `zeroLocus`/`vanishingIdeal`), `Nat.divisorSum` (B; `ArithmeticFunction.sigma`).

The 26 false flags decompose into six mechanical modes: namespace short names standard under `open` (13), self-declared theorem headers (4), dot-notation on a variable (5, e.g. `M.det`), comment prose (2), an import-line module name (1), a mid-dot-chain fragment (1).

Repair rules R1–R5: drop comment tokens, import-line names, self-declarations, and single-letter dot-notation heads; classify a name as existing if some single-segment prefix `NS.` makes it an indexed declaration. Validation: the repaired classifier agrees with the blinded truth on **59/60** (sole miss: the mid-dot-chain fragment `Laurent.coeff.support`).

Run-level rates, both instruments (n=100/arm; citation-level shown for context — citations cluster within runs, so run-level carries the inference):

arm	raw runs ≥1 halluc	raw citation-level	repaired runs ≥1	repaired citation-level	R5 reclassified	R1–R4 dropped
A	54/100 [44.3, 63.4]	94/443 = 21.2%	37/100 [28.2, 46.8]	50/428 = 11.7%	29	15
B	48/100 [38.5, 57.7]	76/429 = 17.7%	30/100 [21.9, 39.6]	38/416 = 9.1%	25	13
C	49/100 [39.4, 58.7]	108/517 = 20.9%	11/100 [6.3, 18.6]	26/500 = 5.2%	65	17
D	23/100 [15.8, 32.2]	32/472 = 6.8%	6/100 [2.8, 12.5]	7/457 = 1.5%	10	15
E	49/100 [39.4, 58.7]	107/505 = 21.2%	10/100 [5.5, 17.4]	14/483 = 2.9%	71	22

Raw run-level McNemar: D-vs-E p=1.6e-4, D-vs-C p=6.9e-5 (run-level ratio 2.13×, not the citation-level 3.1×). Repaired: D-vs-E p=0.454, D-vs-C p=0.302, D-vs-A p=1.23e-7, D-vs-B p=3.9e-5, E-vs-C p=1.0. Commit-clustered bootstrap versions of the significant contrasts (`v3_gate_fixes.json`): D-vs-A −0.310 [−0.446, −0.193] p=0.0002, E-vs-A −0.270 [−0.396, −0.155] p=0.0002; the null D-vs-E / D-vs-C stay null (clustered p=0.44 / 0.31). The R5 column is the citation-style artifact quantified: C/E write namespace-short names (65/71 reclassifications), D copies fully-qualified names out of tool payloads (10).

Held-out revalidation of the repaired classifier (seed 20260802). The five rules were formulated after the 60-name audit unsealed, so 59/60 is an in-sample fit. A second blinded pass (`bench/analysis/halluc_holdout.{py,json,md}`; blinded intermediates in `bench/analysis/holdout_blind/`) sampled **40 distinct cited names disjoint from the 60** (stratified per arm: 4 raw-hallucinated, 3 raw-exists, 1 raw-renamed; shortfalls → exists) from the post-repair fresh rows, graded them blind under the same evidence protocol (raw `git grep` at `61a5e4f338`, cross-check `9944fe2973`, run outputs for self-declaration/variable resolution), and only then unsealed both instruments' verdicts. Truth labels: real 31 · nonexistent 3 · not_a_citation 5 · renamed 1.

metric (strict, n=40)	raw oracle	repaired classifier
flagged-class precision	3/20 = 15.0% [5.2, 36.0]	3/7 = 42.9% [15.8, 75.0]
recall (true fabrications)	3/3 = 100%	3/3 = 100%
accuracy	57.5%	90.0%
binary agreement (the in-sample 59/60 measure)	23/40	**36/40 = 90.0%** [76.9, 96.0]

Sensitivity: reading the borderline `OrderedSemiring` (a class removed by the ordered-algebra refactor; labeled renamed on the Basis→Module.Basis precedent) as nonexistent gives repaired precision 4/7 = 57.1%. The verdict: **directionally replicates** — the raw oracle's invalidity reproduces almost exactly (13.3%→15.0% precision), the repair transfers most of its value (FPs on sampled flags 17→4, accuracy 57.5%→90.0%, zero FNs for either instrument) — but the in-sample fit was optimistic (98.3%→90.0% binary agreement, Fisher p=0.15), and a repaired flag is roughly a coin toss, not a confirmation. The four residual FP modes are ones R1–R5 never saw: multi-segment namespace short names (`IsZero` ← `CategoryTheory.Limits.IsZero`), dot-notation on a namespaced def value (`MeasureTheory.volume.restrict`), a multi-char local variable (`M001`), and the removed-class borderline (`OrderedSemiring`). Implications: the adjusted per-arm rates likely overstate true hallucination by roughly 2× and should be read as **upper bounds**; two of the four residual FP modes are namespace-style-correlated (both held-out instances land in C/E-style short-name arms), so the residual bias still runs *against* the free-text tool arms and cannot manufacture D's advantage; and all three held-out confirmed fabrications (`Subgroup.torsion`, `Localization.map`, `Module.GeneralizedEigenspace`) sit in the no-formal-tools arms A/B, as in-sample.

S4 Fresh-set exposure strata (full)

The source is `bench/analysis/fresh_exposure.{py,json,md}` (original, snapshot basis), recomputed on the post-repair rows by `bench/analysis/v3_gate_fixes.{py,json}` §1. Pinned tree `61a5e4f338` (content ∼2026-07-10), the rev C/E's file tools read. Outcomes below are the **post-repair rows** (E's 31 outage rows replaced by their 2026-07-27 reruns) on the **raw** instrument; the superseded snapshot-basis table (E's 429 rows counted as failures) is preserved in `fresh_exposure.md`.

Exposure counts (of 100): full dotted name in-tree verbatim **37**; basename as a declaration header anywhere **64**; basename as header in the task's own module **51** (the stratum basis); gold's commit an ancestor of the pin **49**. (Two tasks are exposed despite post-pin merge commits — measured on tree bytes, not merge metadata.)

arm	exposed (n=51)	unexposed (n=49)
A	9/51 = 17.6% [9.6, 30.2]	11/49 = 22.4% [13.0, 35.9]
B	8/51 = 15.7% [8.2, 28.0]	14/49 = 28.6% [17.8, 42.4]
C	11/51 = 21.6% [12.5, 34.6]	14/49 = 28.6% [17.8, 42.4]
D	24/51 = 47.1% [34.1, 60.5]	18/49 = 36.7% [24.7, 50.7]
E	15/51 = 29.4% [18.7, 43.0]	15/49 = 30.6% [19.5, 44.5]

McNemar by stratum, post-repair: D-vs-E exposed 14/5 p=0.064, unexposed 11/8 p=0.648; D-vs-C exposed 18/5 p=0.0106, unexposed 10/6 p=0.454. The leak's direction favors C/E. **A correction to earlier editions:** the snapshot-basis version of this table showed D's edge over E strongest in the *unexposed* stratum (17/3, p=0.0026), and v2 concluded that D's advantage was "strongest exactly where there was nothing to leak." That pattern does not survive the E repair — 24 of E's 31 outage rows fell in the unexposed stratum, so the stratum contrast was outage-driven. On the repaired rows D-over-E, like D-over-C, is *larger in the exposed stratum* (RD +0.18 vs +0.06) and is a null unexposed; neither stratum is significant alone. The merge-date split reproduces the repaired pattern (before-pin 14/5 p=0.064; after-pin 11/8 p=0.648), and the repaired-instrument sensitivity gives the same qualitative picture (D-vs-E exposed 15/8 p=0.21; unexposed 13/9 p=0.52). The judge-endpoint exposure split is §S11.4.

S5 Turn-budget sensitivity

The source is `tier1_reanalysis.md` §5, corrected to the post-repair rows by `v3_gate_fixes.py` §4. E overran the advisory 30-turn budget on **48/100** repaired rows — 16 of the 31 rerun rows plus 32 of the 69 originals; the earlier as-run count of 32 treated E's 31 outage rows (turns=1) as within budget, which also inflated the old within-budget count (E 68) and the old n=45 pair analysis. Corrected per-arm within-budget run counts: A 100, B 100, C 50, D 62, E 52. Restricted to pairs where both arms stayed within the advisory 30 turns (raw instrument, post-repair rows): D-vs-E n=35, 51.4% vs 40.0%, 7/3 discordant, p=0.34; D-vs-C n=35, 54.3% vs 25.7%, 11/1, p=0.0063; D-vs-A n=62, 46.8% vs 25.8%, 19/6, p=0.0146. Consistent in direction; for D-vs-E, inconclusive within budget.

S6 Tool-call census, cost/latency, and the tool-use figure

Tier-1 eval-split trace attribution (`bench/trace_analysis.py`): 98–100% of traced tool-arm runs cite at least one declaration that surfaced in a tool result; 35% of arm-D eval runs cite a declaration from a `brain_bridge/brain_transfer` result. Among *hallucinated* citations (raw oracle), the checked-and-cited-anyway rate on eval rows is 35% (C), 30% (E), 13% (D); on the fresh split — the paper's primary evidence base — the same statistic is C 44%, D 34%, E 61%: verify-before-cite discipline transferred imperfectly to the held-out set, and worst in E.

v2 census (Figure S1, regenerated on the repaired run rows; the per-tool counts quoted here are the as-run census — the race repair raises only F's means, since its 190 condemned rows had zero calls; basis stated per clause): W averages 3.5 calls/run on QR-810 and 10.6 on MPR, ≈4.0 pooled (QR-810 counts: decl_exists 1,251 + brain_bridge 608 — verify-then-cite made mechanical; pooled QR+MPR counts: decl_exists ∼1,473 + brain_bridge ∼727); WF 4.6 pooled QR+MPR in a genuine dual-toolkit mix (decl_grep ∼1.5k, decl_exists ∼0.9k, brain_bridge ∼0.7k, decl_read ∼0.5k, loogle ∼0.4k); repaired F averages 2.8 (QR) and 8.2 (MPR) calls/run. Under the manual, `brain_cell` misuse (63.3% failure over 482 W-arm calls) falls to a single call in WF. Same-eval-set caveat as the manual (§S8).

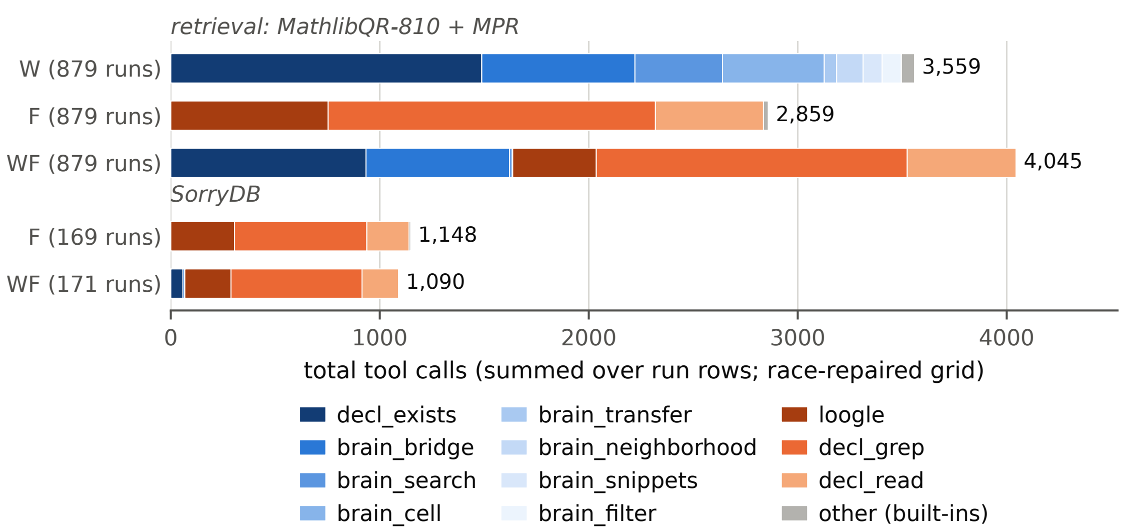

Figure S1: The v2 tool-use census by tool family (Brain blues, formal oranges), summed over run rows — regenerated on the race-repaired grid (fig_f4_tooluse.py; census assertions against the §S6 values inside the script). Arms: W = the Brain; F = formal search; WF = union + manual (post-hoc). The race repair raises only F's totals; the prose census above quotes the as-run figures, basis stated per clause.

The cold-start-race repair (grid provenance). The §3.4 race was found by the U launch and repaired across seven commits. `458891dc` built the U arm (bare W $\cup$ F union, no manual); its first launch showed 13/49 rows with both MCP servers still `pending` at the init event and zero tool calls, and `834a130a` taught the runner to capture that init signature, condemn-and-retry such rows, stagger the first wave, and raise the attach timeout. `99a2075f` byte-archived the 194 affected grid rows (F: 175 QR-810 + 15 MPR; W: 2 + 2; manifest of qids included) to `bench/v2/runs/agent/race_condemned_archive/`; `f041928a` rerun all 194 attach-verified in place under the fixed harness (the 4 stubborn W rows against the original fast-attaching local worker — the asymmetry that had spared W), leaving **zero race rows in every arm × benchmark**; N/U/WF audit race-free before and after (N has no MCP by design), and pre-existing format/timeout failures (N 143+15, F 1, W 5, WF 1) are untouched. `c1dea98f` landed the 879 attach-validated U rows (26 condemned first attempts rerun; $187.20); `322e84f5` and `dd3eb689` are the union-ablation and repaired-grid analyses (`union_ablation.{py,json,md}`, `grid_repaired.{py,json,md}`). Repair cost $33.08. Scores moved only for F (QR R@10 .8309→.8457, nDCG .7901→.8089; MPR gR@10 .4532→.5468); N/U/WF rows are byte-identical and W's aggregates are unchanged at four decimals.

Per-query cost/latency (QR-810) — repaired grid, recomputed from the final run rows (as-run table: `retrieval_repair.md` §3; CLI cost estimates under Max auth):

system	mean $/query	mean wall s	median wall s	mean calls	mean turns
N	0.085	15.1	8.9	0.0	1.0
F	0.143	13.2	10.6	2.8	3.8
W	0.197	23.7	11.2	3.5	4.3
U	0.197	12.4	9.8	3.1	4.1
WF	0.207	15.2	12.5	4.2	5.2
retriever-mode wikibrain	∼0	—	—	1	—

MPR (repaired grid): N $0.185/50.7 s/0 calls · F $0.370/38.8 s/8.2 · W $0.459/62.6 s/10.6 · U $0.403/33.1 s/7.9 · WF $0.418/38.2 s/9.3. U-arm totals: $159.37 (QR-810) + $27.84 (MPR) = $187.20.

N format-repair detail (`retrieval_repair.md`): symmetric tiered extraction (relaxed JSON → back-ticked names → bare dotted identifiers → compound tokens) over all assistant text, identically for every arm's empty rows. QR: N 143 empty rows → 104 repaired ≥1 name but only 12 recover the gold; strict 0.6333 → lenient 0.6481; oracle ceiling 0.8099. F/W/WF have 1/5/1 empty rows on the repaired grid (2/5/1 as-run) and move ≤0.0012. The five W empties are 420 s hard timeouts, not format failures. MPR: N 15 empty → 1 gold recovered (0.2025 → 0.2062). Per-tool provenance detail (surfaced-by-tool, first-written-in, per-style tables): `bench/analysis/retrieval_provenance.md` — an as-run trace pass; its F row counts the 175 race rows' hits as memory (main §5 caveat).

S7 Snapshot rot: the detailed narrative

The original SorryDB freeze (8 repositories, 164 tasks) was prepared against sorrydb.org's SorryDB-2601 evaluation split with pinned upstream commits. At verification time — roughly six months after the pins were minted — 74 of 164 tasks (45%) were unreproducible: the pinned commits of GlimpseOfLean, LeanCourse25, and most Foundation and SemicircleLaw tasks no longer exist on GitHub, their branches force-pushed or rebased away. The rot is invisible to any cheap check: GitHub's archive endpoint answers a preflight HEAD request for a dead commit with HTTP 200, and only an actual fetch fails. The refreeze (`bench/v2/sorrydb_prep.py`, commit `db679e8d`) keeps verified-fetchable pins only — every task's repository archive downloaded and its splice point confirmed — yielding 171 tasks across 10 repositories. Lesson for benchmark builders: a pin is not a snapshot; archive the bytes, not the reference, and verify by fetching, not by HEAD.

S8 The WF agent manual (verbatim) and its timeline

Timeline (verified commit times). N/F/W results committed 22:27 on 2026-07-24; the manual 00:30 on 2026-07-25; WF results 01:15 on 2026-07-25. The manual distills measurements from the same evaluation queries WF was then scored on — its bracketed figures (per-style nDCG, the 63% brain_cell failure rate, the 143/810 format failures, call counts) are the N/F/W/system-mode eval-set aggregates, reproduced exactly.

WF is therefore development evidence of a maximally equipped, briefed agent, not an untouched test result. (v1 also misattributed the 63% figure to “the Tier-1 traces”; it is the v2 W-arm figure — 482 calls, 63.3% failed.)

> `bench/v2/AGENT_MANUAL.md`, prepended verbatim to every WF-arm prompt; a **post-hoc, benchmark-informed** artifact.

Tool manual: searching Mathlib with the Brain + formal search

You have two complementary toolkits. Every claim in this manual is measured from ∼5,500 logged tool calls across 3,500 benchmark runs (Tier-1 + Bridge v2); numbers in brackets are those measurements.

The mental model

- **The Brain (wikibrain, 8 tools)** is a *curated knowledge graph* joining informal mathematics (Wikipedia/Wikidata concepts, external databases) to Mathlib declarations. It knows what things *mean* and how concepts relate — but it only “atomizes” declarations that some concept, annotation, or tag claims (∼12k of Mathlib’s declarations). It is a map, not the territory.
- **Formal search (3 tools)** operates on *all* of Mathlib directly: type-pattern search (loogle), source grep, source reading. It knows everything that exists but nothing about what it means informally.

Rule of thumb: **route by what you’re holding.** Holding a concept described in words → start with the Brain. Holding a type shape or name fragment → start with formal search. Always finish with `decl_exists` on every name you intend to output.

Formal search tools

loogle
Type-pattern and constant search (the public loogle.lean-lang.org service).

- USE for: “a lemma whose statement looks like `_ * _ ≤ _ * _`”, searches by involved constants (`Real.sqrt`, `tsum`), name substrings.
- Strengths: covers ALL of Mathlib; ∼0% call errors [578 calls, 0 errors]; the best premise-finding tool — the loogle+grep toolkit matched a specialist premise retriever [group-recall@10 0.453 vs 0.461 SOTA].
- Weaknesses: needs Lean syntax fluency; a wrong pattern silently returns unrelated hits; small result payloads mean you often need decl_read next.

decl_grep
Regex search over the Mathlib source checkout (ripgrep).

- USE for: name fragments (“cantor”), notation, docstring phrases, finding the file a declaration lives in.
- Strengths: the workhorse [1,206 calls, 0% errors]; catches things loogle’s type index can’t (comments, docstrings, naming conventions).
- Weaknesses: lexical only — if the concept’s name doesn’t appear in source, grep cannot find it (descriptive queries like “special case of X” score worst for grep-based agents [nDCG 0.384 vs the Brain’s 0.523]).

decl_read
Read declaration source (with surrounding context) from the checkout.

- USE for: verifying a candidate actually says what you think BEFORE citing it; getting exact signatures and hypotheses.
- Strengths: ground truth for any decl in Mathlib [397 calls, 0% errors].
- Weaknesses: you need the file/name first — it is a confirmation tool, not a discovery tool.

Brain tools

brain_bridge — THE ENTRY POINT for informal→formal
Free text in, existence-verified declarations + owning atoms out, with signatures, import lines, and one-hop dependency context.

- USE for: “the concept described as ⟨words⟩” — especially nicknames and special-case descriptions, where the Brain’s curated bonds beat grep [special_case nDCG 0.523 vs 0.384; nickname 0.779 vs 0.757].
- CRITICAL LIMITATION (measured): its free-text matching is label/alias anchored, NOT semantic. A single call with a descriptive paraphrase usually misses [single-shot benchmark: 3.6% R@10 vs 82% for an agent that reformulates]. **Never accept one miss as the answer: reformulate 2–4 times** — try the concept’s canonical name, a synonym, the head noun alone. 8.9% of calls return match:“none” [727 calls]; that response includes nearest-candidate suggestions — read them.
- The `hits` list is existence-verified; bond quality matters: `exact` means the decl IS the concept; `formalizes`/weaker bonds are nearby.

brain_search
Fuzzy label + alias search returning atoms (cells).

- USE for: resolving a name you half-know into an atom id; browsing what the Brain calls something [400 calls, 0% errors — the reliable fallback when brain_bridge misses].
- Weaknesses: returns atoms, not ranked decls — follow with brain_cell on the winning atom.

brain_cell

The full atom card: Lean code, Wikidata description, DB snippets, organs.

- USE for: everything known about ONE object after bridge/search resolved it.
- **THE #1 MISUSE [63% of 482 calls failed]: calling it with a bare `decl:Mathlib:Name` for an arbitrary declaration.** The Brain only has atoms for concept-joined decls; "unresolvable key" means *no atom owns this decl*, NOT that the decl doesn't exist. For arbitrary decls use decl_exists (works for all of Mathlib) and decl_read for source. Call brain_cell only with ids that came OUT of a Brain tool.

decl_exists — THE DISCIPLINE TOOL

Batch existence check for declaration names (backed by the full oracle — covers ALL of Mathlib, unlike brain_cell).

- **USE ALWAYS, on every name you are about to output.** This is the measured difference between grounded and hallucinated citations: agents that verified before citing cut hallucinated-but-cited-anyway to 13% vs 33% for agents relying on fuzzy search results alone. It is cheap [1,473 calls, 0.1% errors] and batched — check 10 names in one call.
- Also returns verified-rename suggestions for stale names.

brain_neighborhood

The synapse graph around an atom (depends/links/relates, cursored).

- USE for: "what connects to X" — finding the connecting lemma between two concepts, walking dependencies.
- Weaknesses: needs a valid atom id [24.8% of calls errored — same unresolvable-key trap as brain_cell; resolve the atom first].

brain_transfer / brain_snippets / brain_filter

Narrower tools: label↔decl jumps for KNOWN labels (transfer), stored source snippets (snippets), facet enumeration (filter). High misuse rates when fed non-atom ids [40%/57% errors]. Prefer brain_bridge/brain_search entry; reach for these only on ids the Brain already returned.

Routing playbook

You are holding	Do this
A concept in words	brain_bridge → (miss?) reformulate → brain_search → brain_cell
A special case / nickname	brain_bridge (the Brain's best terrain)
A type shape	loogle
A name fragment	decl_grep
Premises for a proof	loogle + decl_grep FIRST [0.453 vs 0.272], Brain for concept grounding
A candidate name to cite	decl_exists (batch, always) → decl_read if the statement matters
An atom id from any Brain tool	brain_cell / brain_neighborhood freely
A bare decl name needing info	decl_exists + decl_read — NOT brain_cell

Anti-patterns (each one measured in prior runs)

1. **Citing unverified names.** 33% of hallucinated citations came from agents that had search evidence in front of them and cited a wrong name anyway. decl_exists is binary and kills this.
2. **One query, then giving up.** The single-call miss rate on descriptive queries is ∼96%; agents that reformulate reach ∼82%. Iteration IS the algorithm.
3. **brain_cell as a generic decl inspector.** 63% failure rate. Brain atoms ≠ all of Mathlib.
4. **Answering outside the requested format.** 143/810 no-tool runs were scored 0 for exactly this. Re-read the output contract before replying.
5. **Inventing tool names** (e.g. `logue`, `lookie`). Use the names above.

(End of verbatim manual.) Its bracketed measurements are the as-run pre-repair aggregates as WF saw them — e.g. the loogle+grep premise figure it quotes as 0.453 is 0.547 on the repaired grid (§S6) — and are preserved verbatim, not corrected.

S9 SorryDB: full bookkeeping

The frozen split holds n=171 tasks/arm across 10 repositories, and all 203 candidate verdicts are definitive — 183 kernel pass/fail plus 19 unspliceable and 1 verify-timeout that never reached the kernel (commit `9682b3b1`; the 8 curve25519 stragglers verified as 0 proved). `verify.jsonl` carries 2 additional off-frozen N rows verdicted `env_broken` (205 lines total), correctly excluded from the 203.

	N no tools	F formal	WF union+manual (post-hoc)
run rows present	168	169	171
no output (timeout / no lean block)	70	15	11
honest give-up (literal `sorry`)	40	83	86
candidate proofs submitted	58	71	74
proved (kernel)	**2**	**9**	**10**
failed	50	55	57
unspliceable	5	7	7
verify timeout	1	0	0
proved / 171	1.2% [0.3, 4.2]	5.3% [2.8, 9.7]	5.8% [3.2, 10.4]
repo-clustered boot 95% CI	[0.0, 2.7]	[0.0, 10.7]	[0.0, 12.2]
proved / candidates	3.4%	12.7%	13.5%
total cost (USD)	58.97	102.76	108.87
cost per proved	29.48	11.42	10.89
mean wall-clock / task	120 s	198 s	183 s

3 N and 2 F tasks lack run rows (runner losses). Distinct sorries proved: 13; N's are a strict subset of F's and WF's; WF∩F 6, WF-only 4, F-only 3. Per-repo: PersistentDecomp N1/F3/WF3, LeanAPAP 0/4/5, MiscYD 1/2/2, the other seven repos 0 everywhere. WF−F: +0.0058, repo-clustered 95% CI $[+0.0000, +0.0185]$ (the lower endpoint is exactly 0 because WF $\geq$ F in every repo; 34% of resamples show no difference or worse); task-level exact McNemar 4/3, p=1.0. Cost-per-proved is bookkeeping only (unstable at 2/9/10 successes). The published SorryDB agentic best (∼30%) is not comparable: different task subset, different protocol (published agents iterate against the build; ours drafted one-shot).

S10 Per-arm cost tables

Tier-1 campaign means per task (all 471 runs/arm as run, pre-repair; E's 31 error rows cost ≈0): A \$0.034 · B \$0.048 · C \$0.140 · D \$0.121 · E \$0.128; mean wall-clock C 116 s, D 89 s, E 106 s. tokens-to-solve was never computed, so P2 is untested; v1's "cheaper and better" framing remains withdrawn. **Judge pass:** \$81.86 (CLI-reported) for 500 gradings + the 50-item re-grade. **v2 retrieval:** per-query tables in §S6. **SorryDB:** totals in §S9.

S11 Blind LLM-judge grading: full tables

Sources are `bench/analysis/judge_fresh_run.py` and `judge_fresh_summary.{py,json,md}`, with the repaired-leg conjunction from `bench/analysis/conjunction_repaired.{py,json}` (deterministic — no sampling). Judge `claude-sonnet-5`; blind (informal statement + gold with binders + candidate; no arm identity, no tools, empty cwd); 500/500 rows, 0 errors; no-output rows auto-fail. **Uncalibrated; exploratory.** The judge ran before the oracle repair, so the conjunction is reported under both typecheck instruments; the judge leg is identical in both.

S11.1 Fresh-100 rates

Strict equivalence (same proposition, same hypotheses):

arm	k/n	Wilson 95% CI
A	11/100	[6.2, 18.6]
B	9/100	[4.8, 16.2]
C	43/100	[33.7, 52.8]
D	8/100	[4.1, 15.0]
E	46/100	[36.6, 55.7]

Evaluated equivalence (mathematical equivalence, high confidence): A 17/100 [10.9, 25.6] · B 16/100 [10.1, 24.4] · C 51/100 [41.3, 60.6] · D 19/100 [12.5, 27.8] · E 53/100 [43.3, 62.5].

Conjunction (grounded-typecheck ∧ evaluated), both instruments:

arm	raw leg	repaired leg [Wilson 95]
A	8/100	8/100 [4.1, 15.0]
B	9/100	9/100 [4.8, 16.2]
C	16/100	20/100 [13.3, 28.9]
D	15/100	16/100 [10.1, 24.4]
E	18/100	21/100 [14.2, 30.0]

Eight rows flip under the repair (C 4, E 3, D 1 — the repaired-clean typecheck passes that the judge also evaluated equivalent); B/A are essentially untouched, and between-tool parity is preserved.

S11.2 McNemar, fresh-100

Evaluated equivalence: D-vs-E 1/35 p=1.1e-9 · D-vs-C 3/35 p=6.7e-8 · D-vs-A 7/5 p=0.774 · E-vs-A 37/1 p=2.8e-10 · E-vs-B 39/2 p=7.8e-10 · E-vs-C 13/11 p=0.839.

Conjunction, raw leg: D-vs-E 6/9 p=0.607 · D-vs-C 8/9 p=1.0 · D-vs-A 9/2 p=0.065 · E-vs-A 11/1 p=0.0064 · E-vs-B 12/3 p=0.035 · E-vs-C 6/4 p=0.754.

Conjunction, repaired leg (`conjunction_repaired.py`): D-vs-E 5/10 p=0.302 · D-vs-C 8/12 p=0.503 · D-vs-A 10/2 p=0.039 · E-vs-A 14/1 p=0.00098 · E-vs-B 14/2 p=0.0042 · E-vs-C 6/5 p=1.0. The only classification change at α=.05 is D-vs-A ns → sig — a strengthening of tools-versus-none; every between-tool conjunction contrast stays null under both instruments.

Commit-clustered bootstrap versions (44 clusters, B=10,000; `v3_gate_fixes.json`): evaluated equivalence D-vs-E -0.340 $[-0.462, -0.220]$ p=0.0002, E-vs-A $+0.360$ $[+0.233, +0.469]$ p=0.0002; conjunction (repaired leg) D-vs-A $+0.080$ $[+0.011, +0.152]$ p=0.037, E-vs-A $+0.130$ $[+0.055, +0.226]$ p=0.0002, D-vs-E -0.050 $[-0.143, +0.021]$ p=0.247. No classification changes: everything significant unclustered survives clustering, and the nulls stay null.

S11.3 Completed-69 continuity

Evaluated: A 10/69 (.145) · B 9/69 (.130) · C 37/69 (.536) · D 11/69 (.159) · E 37/69 (.536); D-vs-E 0/26 p=3.0e-8; conjunction (raw leg): C 8/69 (.116) · D 10/69 (.145) · E 9/69 (.130), D-vs-E 4/3 p=1.0, D-vs-A 7/0 p=0.016; conjunction (repaired leg): C, D, and E at exactly 11/69 (.159) each, D-vs-E 4/4 p=1.0, D-vs-A 8/0 p=0.0078.

S11.4 Exposure split (evaluated equivalence)

arm	exposed (n=51)	unexposed (n=49)
A	8/51 (.157)	9/49 (.184)
B	9/51 (.176)	7/49 (.143)
C	36/51 (.706)	15/49 (.306)
D	10/51 (.196)	9/49 (.184)
E	35/51 (.686)	18/49 (.367)

D-vs-E exposed: 1/26, p=4.2e-7. D-vs-E unexposed: 0/9, p=0.0039 — the inversion narrows off-leak but does not vanish; E produces more judge-equivalent statements than D even where the gold was not retrievable. The conjunction (S11.2) is where parity appears. This is reported as measured; the human queue below is how it gets adjudicated.

S11.5 Self-consistency and the human queue

Fixed 50-item seed-stratified re-grade (seed 20260727, 10/arm): strict agreement 98.0% (one flip), evaluated agreement 100.0%. Human grading queue = the 36 D/E-discordant tasks on evaluated equivalence (they drive the D-vs-E McNemar): fresh_002, 004, 006, 007, 011, 012, 013, 014, 019, 021, 023, 026, 027, 032, 044, 048, 049, 050, 051, 056, 057, 059, 062, 063, 064, 067, 069, 076, 077, 085, 086, 092, 094, 095, 096, 099.

S12 Responses to the external reviews

Both reviews are preserved verbatim in `docs/research/review/` (REVIEW-1.md, REVIEW-2.md), with our independent claim-by-claim verification of review 1 in `verification-of-review-1.json`.

S12.1 Review 1 (of v1, 2026-07-27) — actions taken in v2

All eight concerns confirmed in substance (29 claims confirmed, 1 partial, 1 refuted). Two factual claims did not survive: MathlibMPR does not cluster by PR (69 tasks, 69 distinct PRs), and arm E's 31 declaration-less runs were contiguous 429 errors, not interface overload — the second correction removed both the review's overload reading and v1's error-inflated headline p=2.4e-5.

#	Concern	Action
1	"Success" measures no semantic correctness	renamed grounded typecheck rate; judge pass; calibration owed
2	D vs E does not isolate the join	"join carries the effect" retired; bundled-condition language; the 2×2 factorial has since been run (v3.1, §4.4/§S15): both main effects null
3	Arm E failed its manipulation check	disclosed; yoked control on the roadmap
4	Fresh set not isolated from formal-search sources	renamed post-Brain-index; exposure strata; frozen snapshots on the roadmap
5	WF is test-set tuned	labeled post-hoc, benchmark-informed everywhere; provenance + timeline (§S8); bare-union arm launched
6	Independence and execution problems	declaration-clustered reanalysis; CIs throughout; disclosures
7	Preregistered experiment incomplete; deviations silent	execution inventory (§S1); register labels; zero-confirmatory up front
8	SorryDB evidence thin	all 203 verdicts completed; WF-vs-F indistinguishable; cost claim deleted

S12.2 Review 2 (of the v2 draft, 2026-08-01) — actions taken in v3

#	Recommendation	Action in v3
1	Lead with repaired full-100; one coherent paired inference	commit-clustered paired bootstrap is the only main-text framework; completed-69 and Wald/McNemar mismatch moved here (§S2)
2	Cluster the fresh set by commit/family	done (`fresh_clustered.py`): 44 commits, 57 files, 59 families; all levels + collapses (§S2.3); D−E widens exactly as predicted
3	Human semantic evaluation	full 500-row blind judge complete with self-consistency; 36-task discordant queue defined; **human grading and calibration still pending** — main §4.2 stays exploratory
4	Factorial ablation	DONE in v3.1: the preregistered 2×2 (prereg `3658bd58`, before any row) is run and analyzed (§4.4/§S15) — join +0.030 p=0.645, verifier +0.050 p=0.230, interaction −0.100 (redundancy); the bare-union U ablation had already shown the union inert (U−F null everywhere), the WF gain manual-driven on QR, and formal tools carrying MPR
5	Characterize the Brain; retrieval provenance decomposition	done (`brain_artifact.py`, main §3.1; `retrieval_provenance.py`, main §5)
6	Real related work (CRAMF, Aria, +)	done; every citation verified against arXiv on 2026-08-01 (`docs/research/review/related_work_notes.md`); comparison table in main §2
7	Correct two overclaims (memorization; "clean verifier finding")	memorization → benchmark-familiarity language (main §4.1); verifier finding restated after blinded oracle validation — between-tool contrast dissolves under the repaired instrument (main §4.3), DDR cited as convergent evidence
8	Repair the retrieval evaluation	symmetric lenient extraction for every arm, cost/calls columns, retriever-vs-agent blocks, Brain-gold coverage, provenance decomposition (main §5, §S6); cold-start-race grid repair + U ablation complete; WF label kept
cuts	move manuals, inventories, strata, response tables, file maps out	this supplement is that move

S13 Changelog

- **v1 (2026-07-25).** Initial report. Headline claims later retired: "the join carries the effect" ($p<0.0001$ driven by an unrecognized outage), "42% success" (no equivalence leg), "contamination-proof fresh set", WF as a SOTA row, cost-per-success claims.
- **v2 (2026-07-31).** Post-review-1 corrected edition: metric renamed grounded typecheck; arm-E 429 block repaired; register labels (preregistered-modified / exploratory / corrective) and zero-confirmatory statement up front; exposure strata; declaration-clustered retrieval statistics; WF relabeled post-hoc; SorryDB verdicts completed; judge and union slots left open.
- **v3 (2026-08-02, this edition).** Post-review-2 restructure into an 8-page paper + this supplement. New evidence: blinded hallucination-oracle validation (13.3% precision) + 5-rule repaired instrument, with all Tier-1 headline numbers recomputed under it; commit-clustered bootstrap as the single inferential framework; completed blind judge pass (500/500) with the three-layer contamination×endpoint analysis; Brain artifact characterization; retrieval provenance decomposition (verification + routing, not content);

N format-failure repair; the cold-start-race grid repair (194 rows rerun attach-verified; F QR R@10 .831→.846, MPR .453→.547) with the bare-union U ablation and final contrast set; the repaired-leg judge conjunction; verified related work; cost columns; conflict/AI-use disclosures. Still open: human grading of the 36-task queue, judge calibration, the 2×2 factorial.

- **v3 post-gate corrections (2026-08-03).** After an adversarial verification gate: held-out blinded revalidation of the repaired oracle (§S3; flags become upper bounds); §S4 exposure strata and §S5 turn budgets recomputed on the post-repair rows (the "strongest-where-nothing-to-leak" claim retracted as outage-driven; E overruns 48/100); commit-clustered bootstraps extended to every significant §4.2/§4.3 contrast; Tier-1 attach audit; fresh-task provenance paragraph; five-arm judge Layer 1; census bases relabeled; assorted number and wording corrections (681, S6 fresh rates, SorryDB verdict categories, anchor provenance).
- **v3.1 (2026-08-18, this edition).** The preregistered 2×2 join × verifier factorial, run and analyzed exactly per `BRIDGE-FACTORIAL.md` (prereg `3658bd58` 2026-08-07; 400 rows `ba35fe7f`; new main-text §4.4 + §S15): both preregistered main effects null (join +0.030 p=0.645, verifier +0.050 p=0.230), exploratory interaction −0.100 (redundancy, not synergy); abstract, §1, §7, and §8 updated to the causal answer. One scoring-phase infrastructure incident (REPL server death mid-D′; the whole arm re-typechecked, §S15.1) documented with full cell provenance. Judge-graded secondaries completed same-day after an authentication delay (400/400 + 40-item re-grade, self-consistency 100%/97.5%): the §4.2 contamination-by-endpoint inversion reproduces (evaluated-equivalence JOIN −0.265 p=0.0002, exploratory; conjunction parity, all effects null; §S15.4). Still open: the 36-task human queue and judge calibration.

S14 Data availability and file map

Everything needed to recompute the report lives in the private preservation repository `Deicyde/wikilean-bridge-experiment` (report at the root; supplement under `report/`). Provenance commits below refer to the WikiLean working repository.

Key commits. `0d36f266` preregistration (2026-07-16) · `f97e67e4` review-1 preservation + byte-preserved 429 archive · `19a90209` arm-E repair driver (MCP-attach detection) · `64c48052` v2 corrective statistics (tier1_reanalysis, fresh_exposure, retrieval_clustered) · `9682b3b1` SorryDB kernel verdicts complete (203/203) · `36f3a472` report v2 + scored summary v2 (`bridge_summary_v2.json`) · `662db420` review-2 analyses (commit-clustered inference, blinded oracle validation, N format repair, Brain artifact table, provenance decomposition) · `3cb8fa55` blind judge grading complete (500/500, self-consistency, conjunction tables, human queue) · `9d10e44f` repaired-oracle Tier-1 recompute (`success_repaired`) · `458891dc` U-arm harness (bare W ∪ F union, no manual) · `834a130a` cold-start-race condemnation + staggered launches · `c1dea98f` U run rows (879, attach-validated) · `322e84f5` union-ablation analysis · `99a2075f` race-row archive (194 rows, byte-preserved) · `f041928a` repaired F/W run rows (zero race rows remain) · `dd3eb689` repaired-grid contrasts + refreshed union ablation · `2a9f6b91` REPL routing fix ("honest labels").

Path (preservation repo)	Contents
`docs/research/BRIDGE-EXPERIMENT.md`	preregistration incl. deviations 1–7
`docs/research/review/`	reviews 1–2 verbatim; verification-of-review-1.json; related_work_notes.md (verified citations)
`bench/*.py`, `bench/arms/`	Tier-1 harness: runner, scorer, REPL rig, trace analysis, arm manifests
`bench/data/bridge_tasks.jsonl`	371 ProofNet# tasks (pinned fork, MIT)
`bench/data/fresh_tasks.jsonl`	100 post-Brain-index tasks + determinacy annotations + holdout check
`bench/data/runs/`	all 2,355 Tier-1 run rows incl. 31 repaired E rows
`bench/data/runs_E_fresh_429_archive/`	the 31 original 429 rows, byte-preserved
`bench/analysis/bridge_summary_v2.json`	scored fresh-500 summary + repair provenance
`bench/analysis/tier1_reanalysis.*`, `fresh_exposure.*`, `retrieval_clustered.*`	v2 corrective statistics
`bench/analysis/fresh_clustered.*`, `halluc_validation.*` (+`halluc_blind/`), `success_repaired.*`	v3: clustered inference; blinded oracle validation; repaired-instrument recompute
`bench/analysis/halluc_holdout.*` (+`holdout_blind/`)	held-out blinded revalidation of the repaired oracle (40 disjoint names, seed 20260802)
`bench/analysis/v3_gate_fixes.{py,json,md}`	post-gate analyses: post-repair S4 strata, clustered §4.2/§4.3 bootstraps, Tier-1 attach audit, corrected turn budgets, five-arm judge table, fresh-task provenance
`bench/analysis/judge_fresh_run.py`, `judge_fresh/`, `judge_fresh_summary.*`	blind judge pass (500 gradings + re-grade)
`bench/analysis/conjunction_repaired.{py,json}`	judge conjunction under both typecheck instruments (deterministic)
`bench/analysis/retrieval_repair.*`, `retrieval_provenance.*`, `brain_artifact.*`	v3: format repair; provenance decomposition (as-run trace pass); artifact characterization
`bench/analysis/union_ablation.*`, `grid_repaired.*`	U-arm bare-union ablation; race-repaired grid, before/after tables + final contrast set
`bench/v2/runs/agent/race_condemned_archive/`	the 194 original cold-start-race rows, byte-preserved with qid manifest
`bench/v2/` + `bench/v2/data/` + `bench/v2/runs/`	v2 harness, AGENT_MANUAL.md, MathlibQR/MPR (CC BY 4.0), SorryDB freeze, all v2 run rows (repaired grid) with gzipped stream transcripts, verify.jsonl (203 verdicts)
`docs/research/BRIDGE-FACTORIAL.md`	the 2×2 factorial preregistration (commit `3658bd58`, byte-unchanged; empty deviations log)
`bench/factorial/`	factorial runner (`run_factorial.py`) + prereg-time live-index census
`bench/arms/mcp-{Ep,X,J,Dp}.json`	the four factorial arm MCP configs
`bench/data/runs_factorial/`	all 400 factorial rows + gzipped stream transcripts + `conditions.json` (commit `ba35fe7f`)
`bench/analysis/factorial_scored.{py,json,md}`	Stage-1 mechanical scoring (repaired+raw oracle, fresh-pin typecheck, integrity gate, `retc_provenance`)
`bench/analysis/judge_factorial_run.py` + `judge_factorial/`	Stage-2 blinded-judge driver + all 400 verdicts + 40-item consistency re-grade (§S15.4)
`bench/analysis/factorial_analysis.{py,json,md}`	Stage-3 preregistered analysis (effects, pairwise grid, sensitivity cuts)

Recomputation entry points. Factorial `bench/analysis/factorial_scored.py` + `factorial_analysis.py` (runner `bench/factorial/run_factorial.py`); Tier-1 grading `bench/score_bridge.py`; repaired instrument `bench/analysis/halluc_validation.py (adjusted)` + `success_repaired.py` + held-out revalidation `halluc_holdout.py`; post-gate analyses `v3_gate_fixes.py`; clustered inference `fresh_clustered.py`; exposure `fresh_exposure.py`; judge `judge_fresh_summary.py` + `conjunction_repaired.py`; retrieval `bench/v2/score_retrieval.py` + `retrieval_clustered.py` + `retrieval_repair.py` + `retrieval_provenance.py`; repaired grid + union ablation `grid_repaired.py` + `union_ablation.py`; artifact `brain_artifact.py`; SorryDB `bench/v2/verify_sorrydb.py`.

External sources (fetched and verified during the v2/v3 design sweeps): TheoremGraph arXiv:2606.25363 · LeanSearch arXiv:2403.13310 · LeanSearch-v2 arXiv:2605.13137 + github.com/frenzymath/LeanSearch-v2 · LeanExplore arXiv:2506.11085 · SorryDB arXiv:2603.02668 + sorrydb.org · FATE arXiv:2511.02872 · miniCTX arXiv:2408.03350 · LeanDojo arXiv:2306.15626 · LeanAgent arXiv:2410.06209 · Numina-Lean-Agent arXiv:2601.14027 · miniF2F-v2 arXiv:2511.03108 · ProofNet#/ProofNetVerif arXiv:2406.07222 · benchmark-faults survey arXiv:2606.29493 · CRAMF arXiv:2508.06931 · Aria arXiv:2510.04520 · DRIFT arXiv:2510.10815 · DDR arXiv:2511.11990 (the last four verified 2026-08-01, `related_work_notes.md`).

S15 The preregistered 2×2 factorial (join × verifier): full detail

Preregistration `docs/research/BRIDGE-FACTORIAL.md`, commit `3658bd58` (2026-08-07) — committed before the harness was built and before any row ran; the file is byte-unchanged since, and its §9 deviations log is empty. Harness commit `cbdcf3b7`; the 400 run rows (+ gzipped stream transcripts + `conditions.json`) commit `ba35fe7f`. Scoring, judging, and analysis are a separate later phase (prereg §4.9), by a different agent from the runner. Artifacts: `bench/analysis/factorial_scored.*` (Stage 1), `bench/analysis/judge_factorial_run.py` + `judge_factorial/` (Stage 2), `bench/analysis/factorial_analysis.*` (Stage 3).

S15.1 Arms, execution, and integrity

2×2 over JOIN (Wikibrain joined surface replaces the unjoined wiki+formal stack) × VERIFIER (`decl_exists` present): E′ = arm E's 6 tools; X = E′ + `decl_exists` (7); J = the 7 Wikibrain graph/content tools; D′ = all 8 (arm D exactly). All arms: `-tools ""` (empty built-in set), per-row allowed/disallowed manifests, model claude-haiku-4-5-20251001, CLI 2.1.153, prompt byte-identical to Tier-1, `-max-turns 30` (mechanical), formal-search reads pinned to the 61a5e4f338 tree, one interleaved order (seed 20260803 shuffle of all 400 pairs), concurrency 4 with staggered first wave, per-row attach-signature + manifest validation with auto-condemn/retry.

Run 2026-08-08 00:51–03:45 UTC, $40.91. Integrity gate (re-verified at scoring): 400/400 terminal rows, 0 errors, 0 attach-dirty, 0 manifest mismatches, 0 zero-tool rows, 0 turn-cap violations; per-arm condition hashes uniform and equal to `conditions.json`. 46 rows hit the mechanical cap (E′ 18, X 22, J 0, D′ 6) — valid terminal rows scored on their extracted output per prereg §4.2.

Scoring-phase incident (defect-4 class, repaired). The fresh-pin REPL server died mid-pass while typechecking arm D′ — during `fresh_052`'s check, whose first verdict recorded the REPL dying — and the 44 subsequent produced rows silently fell back to single-shot bare-environment checks (sub-second failures citing unknown identifiers) — the exact silent-fallback mode the v3 report's §3.4 defect 4 documents. Caught by an elapsed-time audit (all legitimate server verdicts take ≥3 s; the other three arms' minima are 3.0–9.8 s), repaired by re-typechecking every produced D′ row against a restarted, identity-gated server: all 48 healthy-window verdicts reproduced exactly; 23 of the 45 affected verdicts flipped, all False→True (`fresh_052` among them). Full cell-level provenance: `factorial_scored.json` → `retc_provenance`.

S15.2 Primary endpoint: grounded typecheck (repaired oracle)

arm	cell	k/100	rate	Wilson 95%	raw oracle	typecheck ok	produced
E′	join−, ver−	31	.310	[.228, .406]	26	33	82
X	join−, ver+	41	.410	[.319, .508]	31	42	78
J	join+, ver−	39	.390	[.300, .488]	36	39	100
D′	join+, ver+	39	.390	[.300, .488]	39	40	93

Preregistered effects (commit-clustered paired bootstrap, 44 clusters, B=10,000, seed 20260803, `cluster_boot_rd` verbatim; α=.05 two-sided):

effect	register	RD	95% CI	p	verdict
JOIN main	confirmatory	+0.030	[−0.093, +0.139]	0.645	null
VERIFIER main	confirmatory	+0.050	[−0.029, +0.121]	0.230	null
interaction	exploratory	−0.100	[−0.245, +0.047]	0.196	no detectable interaction

Six pairwise clustered RDs (supporting descriptives): D′−E′ +0.080 [−0.071, +0.207] p=0.310 · D′−X −0.020 [−0.172, +0.120] p=0.840 · D′−J +0.000 [−0.099, +0.080] p=1.000 · X−E′ +0.100 [−0.022, +0.215] p=0.127 · J−E′ +0.080 [−0.053, +0.202] p=0.259 · J−X −0.020 [−0.155, +0.116] p=0.821.

Raw-oracle sensitivity: rates E′ 26 / X 31 / J 36 / D′ 39; join +0.090 [−0.006, +0.183] p=0.076; verifier +0.040 [−0.030, +0.103] p=0.298; interaction −0.020 [−0.156, +0.134] p=0.832. The raw instrument flatters the join arms (they copy fully-qualified names from tool payloads; §4.3's citation-style artifact), and the repair removes exactly that artifact — the preregistered primary is the repaired row.

S15.3 Secondary: run-level repaired hallucination (lower better)

Rates: E′ 9/100 · X 5/100 · J 11/100 · D′ 10/100. Effects: join +0.035 [−0.026, +0.100] p=0.289; verifier −0.025 [−0.081, +0.025] p=0.381; interaction +0.030 [−0.082, +0.142] p=0.666. With formal tools of any kind present, flagged citations no longer separate arms.

S15.4 Secondaries: judge evaluated-equivalence and the conjunction

Graded 2026-08-18, after a same-day authentication delay (the judging CLI's Max OAuth had expired; interactive re-login, then the staged pass ran unchanged): `bench/analysis/judge_factorial_run.py`, protocol identical to the v3 fresh-set judge pass (verbatim `judge_bridge.PROMPT`, gold with binders, arm-substring blinding scan green over all 353 gradeable outputs, no-tools claude-sonnet-5 from an empty scratch cwd, `-max-turns 1`, concurrency 3, 429 wait-retry); the 47 no-output rows pre-decided not-equivalent by definition. 400/400 verdicts, 0 judge errors (three transient max-turns failures re-graded cleanly); judge cost $52.32. Self-consistency on the fixed 40-item seed-stratified re-grade (seed 20260727, 10/arm): evaluated agreement 100.0%, strict agreement 97.5% (one flip). Both endpoints are preregistered-exploratory; the judge remains uncalibrated (§4.2 caveats apply unchanged).

Evaluated equivalence:

arm	evaluated	Wilson 95%	strict	exposed (n=51)	unexposed (n=49)
E′	46/100	[36.6, 55.7]	39/100	30/51 (.588)	16/49 (.327)
X	42/100	[32.8, 51.8]	36/100	34/51 (.667)	8/49 (.163)
J	20/100	[13.3, 28.9]	13/100	9/51 (.176)	11/49 (.224)
D′	15/100	[9.3, 23.3]	12/100	7/51 (.137)	8/49 (.163)

Factorial effects on evaluated (same clustered machinery): join −**0.265** [−0.368, −0.154] p=0.0002; verifier −0.045 [−0.117, +0.029] p=0.264; interaction −0.010 [−0.152, +0.117] p=0.915. This is §4.2's contamination-by-endpoint inversion reproduced under the factorial's yoked interface: the unjoined arms hold source grep over the checkout containing 51 of the 100 golds and transcribe — their judged-equivalence advantage concentrates on the exposed half and largely vanishes off it — while the join arms' generate-and-verify outputs are grounded but less often judged equivalent.

Conjunction (grounded typecheck ∧ evaluated, the closest faithful@budget analogue): E′ 20 / X 25 / J 17 / D′ 14 per 100; join −0.070 [−0.161, +0.021] p=0.141; verifier +0.010 [−0.053, +0.065] p=0.822; interaction −0.080 [−0.176, +0.011] p=0.108 — parity: no factor moves the conjunction detectably in either direction.

S15.5 Manipulation checks, descriptives, sensitivity cuts

Verifier usage: X 213 `decl_exists` calls across 94/100 runs; D′ 536 across 92/100. Informal-tool touches: E′ 2/100 runs, X 4/100 — the unjoined arms again barely open the informal corpus (Tier-1's E manipulation failure, reproduced). Mean assistant turns: E′ 20.4 / X 21.4 / J 17.1 / D′ 17.4; mean tool calls 30.4 / 30.3 / 26.8 / 26.7. Production: E′ 82 / X 78 / J 100 / D′ 93 of 100 — the join arms finish inside the 30-turn budget; the unjoined arms cap out (18 and 22 capped rows). The factorial's rates are therefore not comparable to the Tier-1 five-arm table (§4.1): the yoked interface (empty built-ins + mechanical cap + interleaving) is stricter, and D′ lands at 39% where Tier-1's D scored 48%.

Sensitivity cuts (primary outcome, exploratory; same machinery):

cut	E′	X	J	D′	join RD [CI] p	verifier RD [CI] p
own-module exposed (n=51)	17	25	20	18	−0.039 [−0.205, +0.108] 0.653	+0.059 [−0.060, +0.170] 0.391
unexposed (n=49)	14	16	19	21	+0.102 [−0.076, +0.243] 0.271	+0.041 [−0.054, +0.130] 0.438
drop 3 live-index leaks (n=97)	31	39	38	38	+0.031 [−0.093, +0.142] 0.632	+0.041 [−0.039, +0.114] 0.353
both-annotator determinate (n=74)	25	35	30	30	+0.000 [−0.156, +0.139] 1.000	+0.068 [−0.013, +0.151] 0.110

Every cut preserves both nulls. The exposed/unexposed join reversal (−0.04 vs +0.10, both null) is the direction §4.2's contamination analysis predicts — the unjoined arms benefit where golds are retrievable — but neither stratum reaches significance.

S15.6 Recomputation

`python3 bench/analysis/factorial_scored.py` (score → retc-arm Dp → judge-summary) then `python3 bench/analysis/factorial_analysis.py`; runner `bench/factorial/run_factorial.py`; prereg-time live-index census `bench/factorial/live_declexists_census.json` (3/100 golds resolvable in the live Brain index at prereg time — the drop-3 sensitivity row above).